

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8639**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	605	Seat No:	2022300109
Course Section:	1				

L_25_108_986_1_1505_41696



Name of the candidate. Shaikh Arab

Candidate's UID number:

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Examination class room number:

5	0	8
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Invigilator's signature with date:

<u>apwla</u> - 08/10/25

(To be filled in by the candidate)
EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)
PROGRAM: COMPS
DAY AND DATE: Wed, 08/10/25
COURSE NAME: DL

SEMESTER :
I / II / III / IV / V / VI / VII / VIII ✓

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

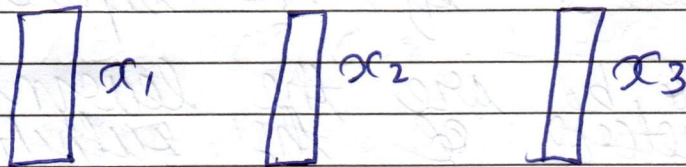
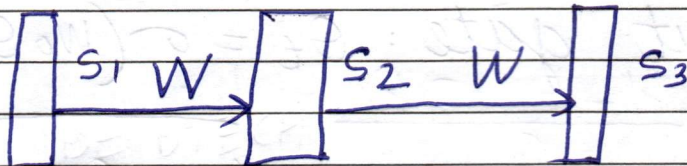
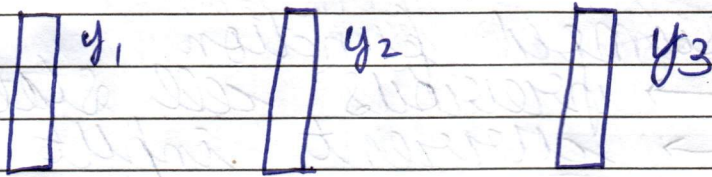
- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.3a Assume that the RNN consists of 3 timesteps with states s_1, s_2, s_3 .

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial s_3} \frac{\partial s_3}{\partial W}$$



$$\text{Now, } \frac{\partial s_3}{\partial W} = \underbrace{\frac{\partial^+ s_3}{\partial W}}_{\text{explicit}} + \underbrace{\frac{\partial s_3}{\partial s_2} \frac{\partial s_2}{\partial W}}_{\text{implicit}}$$

$$\therefore \frac{\partial s_3}{\partial W} = \frac{\partial^+ s_3}{\partial W} + \frac{\partial s_3}{\partial s_2} \frac{\partial^+ s_2}{\partial W} + \frac{\partial s_3}{\partial s_2} \frac{\partial s_2}{\partial s_1} \frac{\partial s_1}{\partial W}$$

$$= \frac{\partial^+ s_3}{\partial W} + \frac{\partial s_3}{\partial s_2} \frac{\partial^+ s_2}{\partial W} + \frac{\partial s_3}{\partial s_1} \frac{\partial^+ s_1}{\partial W}$$

$$\therefore \frac{\partial s_3}{\partial W} = \sum_{i=1}^3 \frac{\partial s_3}{\partial s_i} \left(\frac{\partial^+ s_i}{\partial W} \right)$$

$$\therefore \frac{\partial s_t}{\partial W} = \sum_{i=1}^t \frac{\partial s_t}{\partial s_i} \left(\frac{\partial^+ s_i}{\partial W} \right)$$

$$\therefore \frac{\partial L}{\partial W} = \frac{\partial L}{\partial s_t} \times \sum_{i=1}^t \frac{\partial s_t}{\partial s_i} \left(\frac{\partial^+ s_i}{\partial W} \right)$$

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Q36

GRUs consist of an input gate & an output gate.

Input gate: $i_t = \sigma(W_i S_{t-1} + U_i x_t + b_i)$

$\sigma \rightarrow$ sigmoid function.

$S_{t-1} \rightarrow$ previous cell state.

$x_t \rightarrow$ current input.

Output gate: $O_t = \sigma(W_o S_{t-1} + U_o x_t + b_o)$

W_i, U_i & b_i are the learning parameters of the input gate.

W_o, U_o & b_o are the learning parameters of the output gate.

In LSTM, apart from an input & output gate, there is a forget state which is used to remove the information from the cell state.

In a GRU, the input & forget gates are tied together.

LSTM consist of a cell state & a hidden state which store long term & short term memory respectively. However, GRUs only consist of the cell state storing the long term memory.

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The states of GRU:

$$\tilde{S}_t = \tanh(W S_{t-1} + U x_t + b)$$

$$\tilde{S}_t = \tanh(W(\theta_t \odot S_{t-1}) + U x_t + b)$$

$$S_t = (1 - i_t) \odot S_{t-1} + i_t \odot \tilde{S}_t$$

For an LSTM,

gates: $i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i)$

$$\theta_t = \sigma(W_o h_{t-1} + U_o x_t + b_o)$$

$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$$

States: $\tilde{S}_t = \tanh(W h_{t-1} + U x_t + b)$

$$S_t = f_t \odot S_{t-1} + i_t \odot \tilde{S}_t$$

$$h_t = \theta_t \odot S_t$$

Thus, it can be mathematically observed that GRUs have fewer parameters compared to LSTMs.



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Q.20 For a standard 2D convolution, assume that the input is an $H \times W$ matrix & the filter is of size $F \times F$.

The output feature map will have the dimensions W_1 & H_1 as a $H_1 \times W_1$ matrix where,

$$W_1 = \frac{W - F + 2P}{S} + 1$$

$$H_1 = \frac{H - F + 2P}{S} + 1$$

Assuming padding $(P) = 0$ & stride $(S) = 1$.

$$W_1 = W - F + 1 \quad \& \quad H_1 = H - F + 1$$

Thus, the computational complexity is $(H - F + 1) \times (W - F + 1)$

For a depthwise separable convolution, the output feature map would be of the size $H_1 \times W_1 \times C_1$,

$C_1 = K =$ no. of filters.

$$H_1 = H - F + 1 \quad \& \quad W_1 = W - F + 1$$

assuming $P = 0$ & $S = 1$.

Thus, the computational complexity is $(H - F + 1) \times (W - F + 1) \times K$

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In a CNN, only a few cells of the matrix are activated during a convolution operation as opposed to the feedforward neural network where all the neurons are activated. This ensures that lesser no. of parameters are trained, which is a critical requirement of image processing.

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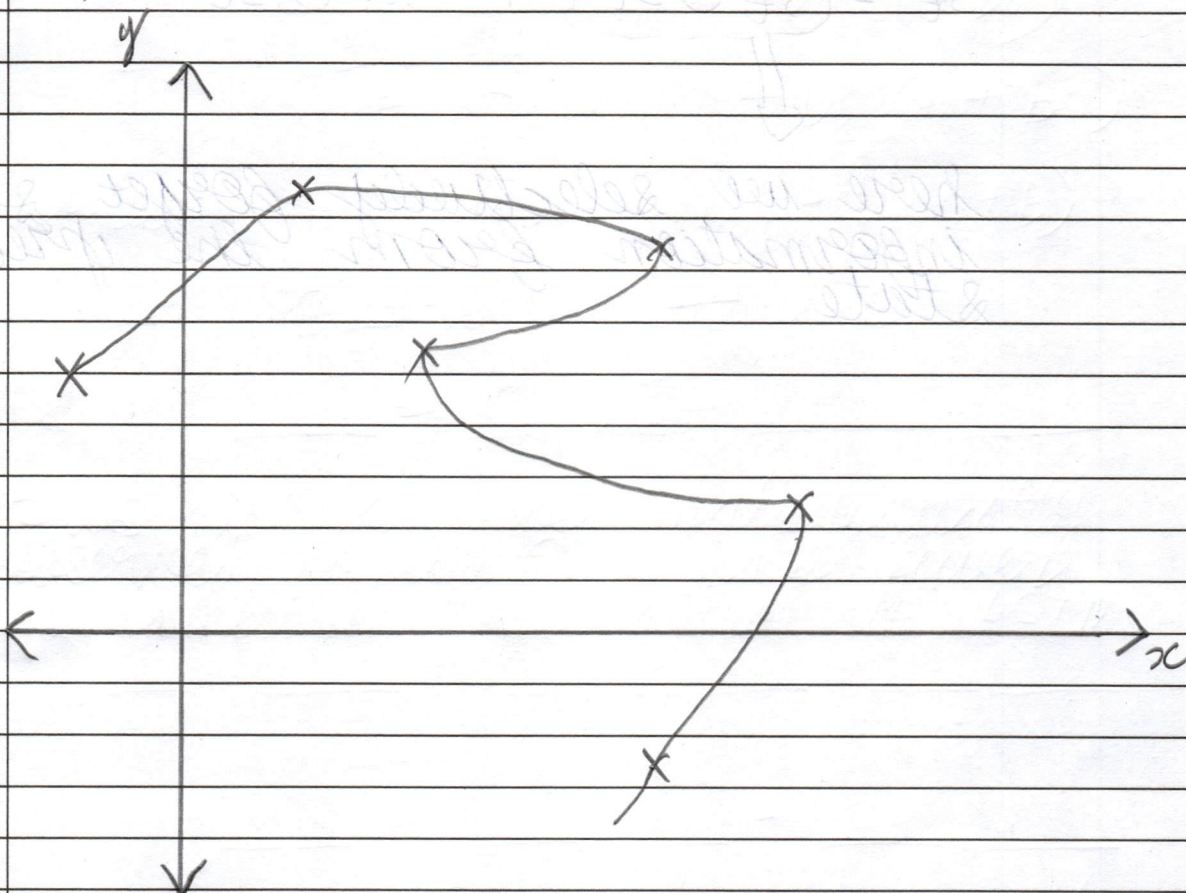
Q.26 Following are the requirements for an autoencoder to be equivalent to a PCA:

- (i) use a linear encoder
- (ii) use a linear decoder
- (iii) use MSE as the loss function
- (iv) normalise the inputs.

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Q.1b When the model capacity is too high, it tends to exactly fit all the patterns in the training data leading to a training error of close to zero.

However, when evaluated on a validation set, the model shows poor performance as it is unable to generalise to the new data. Thus, a case of overfitting is observed.



A model with higher degree learning all patterns leads to overfitting

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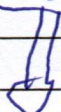
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Q.3a → continued.

The forget gate in an LSTM is given as follows:

$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$$

It is used to remove the irrelevant information from the long-term memory & store only the useful information.

$$S_t = (f_t \odot S_{t-1}) + i_t \odot \tilde{S}_t$$


here we selectively forget some information from the previous state.

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Q.1a The gradient update rule for gradient descent is as follows:

$$W_t = W_t - \eta \nabla W_t$$

$$b_t = b_t - \eta \nabla b_t$$

where, $\eta \rightarrow$ learning rate
 $\nabla W_t \rightarrow \frac{\partial(L(\theta))}{\partial W_t}$

$$\nabla b_t \rightarrow \frac{\partial(L(\theta))}{\partial b_t}$$

The loss function is given by,

$$L = \frac{1}{N} \sum_{i=1}^N (f(w) - y)^2$$

$$= \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{2} (wx - y)^2 - y \right)^2$$

∇W_t & ∇b_t are calculated with the help of back-propagation

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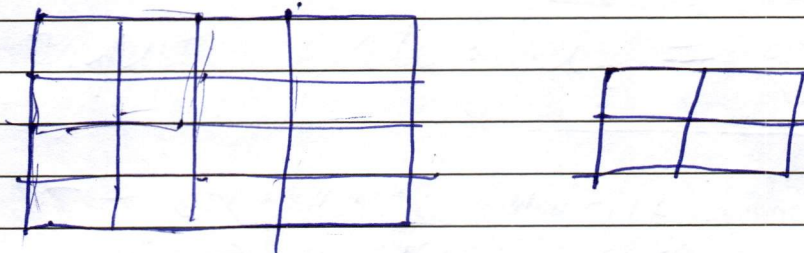
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(a)

(b)